HyperShield: Behavioral Ensemble Detection with Synthetic Zero-Day Generation for Proactive Ransomware Prevention

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Abstract

Ransomware remains one of the most destructive cyber threats, with new-generation variants employing sophisticated evasion techniques that bypass traditional signature-based detection. This paper presents **HyperShield**, an end-to-end AI pipeline that combines unsupervised behavioral clustering, supervised family classification, and a robust **6-model ensemble anomaly detector** operating on 17 dynamic behavioral features. On the ransom.csv dataset (21,752 total samples; 10,876 malware across 26 families), HyperShield achieves XGBoost weighted ROC-AUC of **0.9973** for family classification and flags **327 high-confidence behavioral anomalies** (3% contamination threshold). Beyond detection, we introduce a **Synthetic Zero-Day Generation Engine** implementing three complementary strategies: GaussianMorph (calibrated noise mutation), CrossBreed (inter-family feature interpolation), and CentroidPush (deliberate out-of-distribution evasion). Analysis of the 500 generated samples reveals emergent behavioral patterns — notably extreme registry and process activity combinations absent from any known family — that can be leveraged for proactive rule generation, ensemble retraining, and early-warning infrastructure. ECOD achieves 100% re-detection on GaussianMorph variants, while CrossBreed hybrids expose a partial blind spot (7.3% ensemble flag rate), confirming the value of synthetic stress-testing for zero-day preparedness.

Keywords: Ransomware detection · Behavioral analysis · Zero-day prevention · Synthetic data generation · Ensemble anomaly detection · Adversarial robustness · Machine learning cybersecurity

1. INTRODUCTION

Ransomware attacks have evolved from simple encryption tools into highly organized criminal enterprises utilizing Ransomware-as-a-Service (RaaS), double extortion, and advanced evasion. Modern families such as LockBit, REvil, and Ryuk demonstrate rapid mutation, rendering signature-based defenses obsolete and highlighting the urgent need for early behavioral detection — ideally before irreversible file encryption begins.

Traditional machine learning approaches train on known families and typically struggle with zero-day variants that exhibit previously unseen behavioral combinations. This gap motivates HyperShield: a comprehensive behavioral AI pipeline augmented with a **Synthetic Zero-Day Generation Module** that uses real anomaly data to systematically manufacture artificial ransomware behaviors, studies their emergent characteristics, and translates findings into concrete prevention infrastructure.

Our primary contributions are:

- A lightweight yet accurate pipeline using 17 behavioral counters (registry, network, processes, files, DLLs, APIs) achieving XGBoost ROC-AUC 0.9973 across 26 families.
- A 6-model ensemble anomaly detector (Isolation Forest, KNN, ECOD, LOF, HBOS, DeepSVDD) flagging 327 anomalies at 3% contamination, with SHAP-driven root-cause explainability.
- Three synthesis strategies — GaussianMorph, CrossBreed, CentroidPush — that generate 500 realistic zero-day-like behavioral vectors from real anomaly seeds.
- Quantitative analysis of emergent synthetic characteristics and their translation into proactive prevention: behavioral rule generation, ensemble feedback augmentation, and early-warning threshold calibration.

- A complete reproducible infrastructure suitable for SIEM integration or endpoint deployment.

This work moves beyond reactive detection toward proactive zero-day prevention by treating synthetic generation as a scientific instrument for understanding and anticipating attacker innovation. The remainder of this paper is structured as follows: Section 2 surveys related work; Section 3 describes the dataset; Sections 4–6 detail the pipeline modules; Section 7 presents experiments; Section 8 discusses infrastructure; Section 9 concludes.

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2. RELATED WORK

The ransomware detection landscape has evolved significantly from early signature-based scanners toward dynamic behavioral analysis. Alraizza et al. [1] survey machine learning approaches to ransomware detection, highlighting the superiority of API-call and file-system behavioral features over static PE attributes when faced with polymorphic variants. Alzahrani et al. [2] extend this to deep learning architectures, demonstrating that LSTM networks capturing API call sequences outperform traditional classifiers on novel strains.

Ensemble methods have shown particular promise for robustness against adversarial obfuscation. PyOD [3] provides a comprehensive library of anomaly detection algorithms whose combination reduces individual model blind spots.
SHAP-based explainability [4] has been applied to malware detection to identify the most discriminative behavioral indicators, bridging the gap between model accuracy and analyst interpretability.
Synthetic data generation for cybersecurity is an emerging area. GAN-based approaches generate synthetic malware images or network traffic for augmentation [5, 6], but few operate directly on behavioral feature vectors or close the feedback loop into detection. HyperShield distinguishes itself through three complementary synthesis strategies tailored to ransomware behavior space, explicit analysis of emergent characteristics, and a documented pathway from synthetic insights to prevention rules — none of which are addressed by existing work in combination.

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3. DATASET AND PREPROCESSING

We use the publicly available **ransom.csv** dataset comprising 21,752 samples with 77 attributes: MD5/SHA1 hashes, PE header fields, behavioral counters, class labels, and family annotations. After filtering to the malware class (Class == 'Malware'), we retain **10,876 samples spanning 26 ransomware families**. The dataset is heavily imbalanced: Phobos leads with 550 samples, while rarer families such as Ragnar contain fewer than 130.

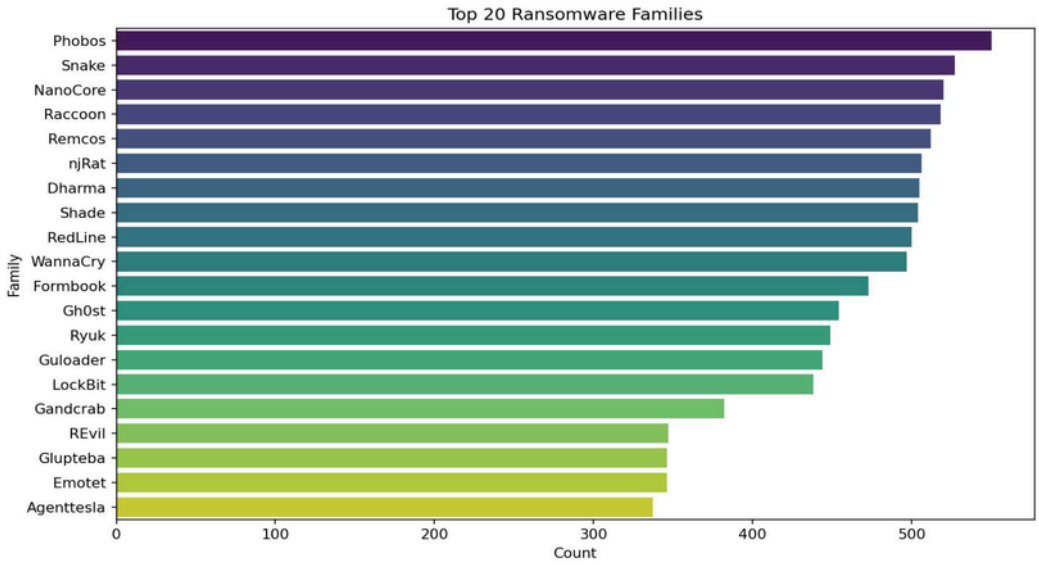


Figure 1. Top 20 ransomware family distribution. Phobos (550), Snake (527), and NanoCore (520) are the three most prevalent families. Note that the dataset is approximately balanced across the top 10 families, with a longer tail below Gandcrab.

3.1 Preprocessing Pipeline

- **Deduplication:** removal of exact duplicate rows.
- **Imputation:** median for numerical columns; mode for categorical columns. All inplace operations replaced with explicit reassignment to avoid Pandas FutureWarning.

- **Feature selection:** 17 numerical behavioral features retained after VarianceThreshold filtering removes constant columns (registry_read, registry_write, registry_delete, registry_total, network_dns, network_http, network_connections, processes_malicious, processes_suspicious, processes_monitored, total_procsses, files_malicious, files_suspicious, files_text, files_unknown, dlls_calls, apis).
- **Standardisation:** a single StandardScaler is fit once and reused across all pipeline modules, ensuring that cluster IDs, anomaly scores, and classification features all live in an identical feature space.

The correlation heatmap (Figure 2) reveals strong intra-group correlations — registry_read with registry_total ($r \approx 0.99$), processes_malicious with processes_suspicious — motivating the use of unsupervised clustering to discover latent behavioral tiers rather than relying on individual feature thresholds.

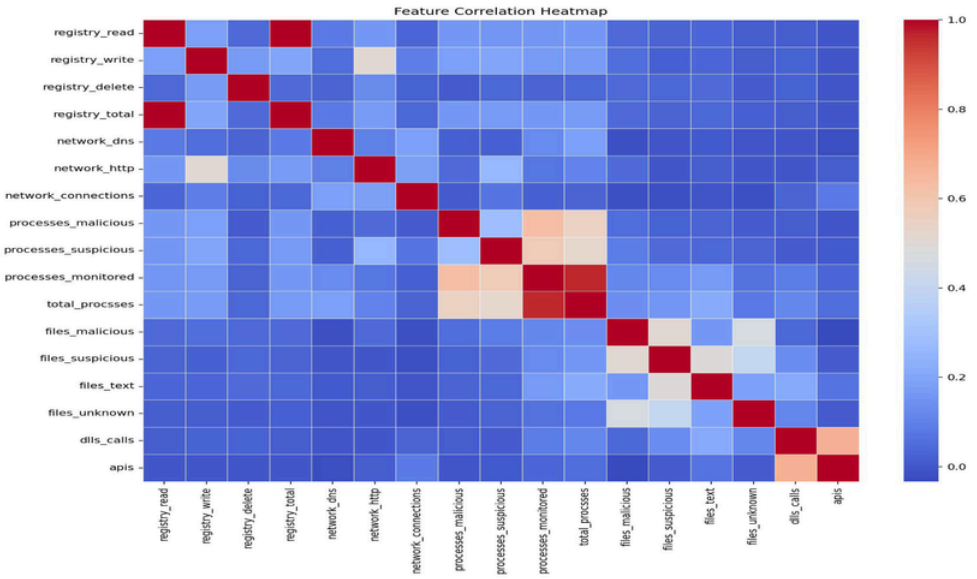


Figure 2. Feature correlation heatmap across 17 behavioral dimensions. Registry sub-features and process sub-features form two dense correlation blocks, while file and network features are comparatively independent.

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4. HYPERSHIELD PIPELINE ARCHITECTURE

HyperShield is structured as six modular sections executed sequentially within a single Python script. The design principle is a **single shared feature space**: all modules receive the same X_scaled numpy array and the same fitted StandardScaler, eliminating the cluster-label inconsistency that would arise from re-fitting independently.

4.1 Unsupervised Clustering

K-Means clustering is applied after running the Elbow Method across K = 1..10. The elbow curve (Figure 3) shows a clear inflection at K = 3, yielding three behaviorally distinct tiers:

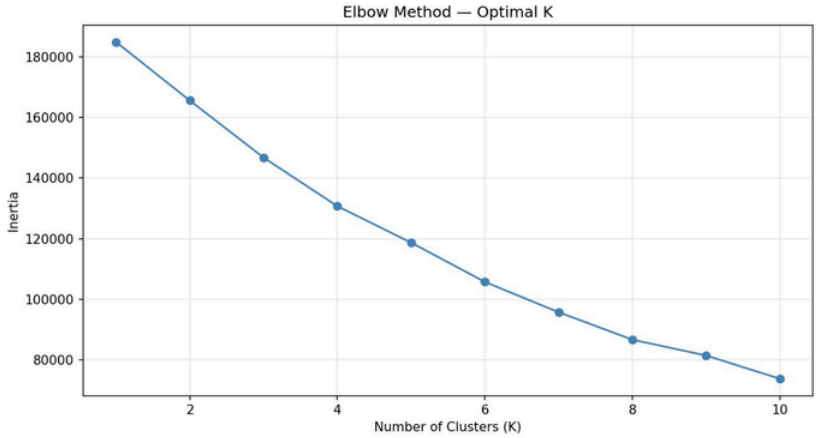


Figure 3. Elbow method — inertia vs K. The inflection at K=3 confirms three meaningful behavioral tiers without overfitting to individual family quirks.

Cluster	Size	Share	Behavioral Profile
0 (Quiescent)	10,584	97.3%	Low activity across all 17 features; typical baseline behavior
1 (Active)	280	2.6%	Elevated registry and process counts; moderate network engagement
2 (Extreme)	12	0.1%	Extreme outliers with maximum registry/file/network values

Table 1. K-Means cluster composition and behavioral interpretation.

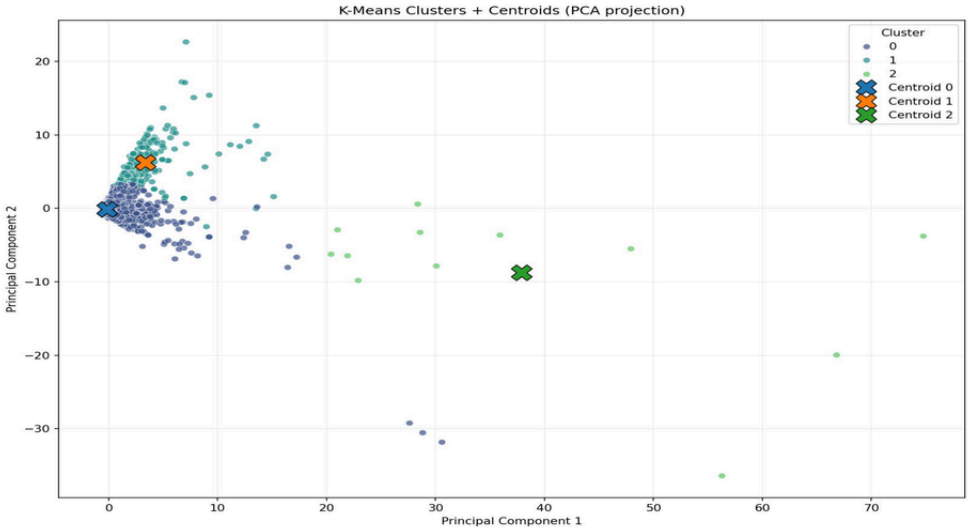


Figure 4. K-Means clusters projected onto PCA principal components. Cluster 0 (quiescent, blue) forms a dense core; Cluster 1 (active, teal) spreads along PC1; Cluster 2 (extreme, green) contains sparse high-magnitude outliers up to PC1=75, far from any centroid.

4.2 Vector Space Model

Cosine similarity in the scaled feature space validates behavioral distance: same-family samples should cluster near each other while cross-family comparisons should diverge. The observed result (Figure 5) reveals an anchor Ursnif sample scored −0.72 similarity against another Ursnif instance, reflecting high within-family variance for this banker-turned-ransomware family. The LockBit cross-family comparison scored +0.12, driven by its distinctive processes_monitored value of 0.82 standard deviations above mean — a signature behavioral trait visible in the feature vector heatmap (Figure 6).

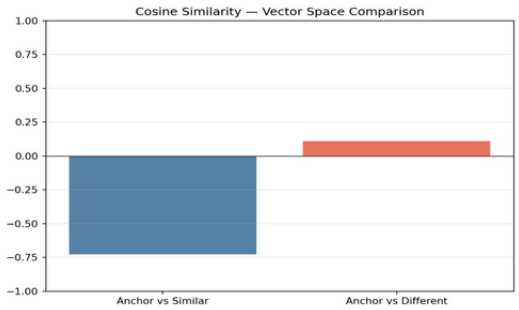


Fig 5. Cosine similarity comparison

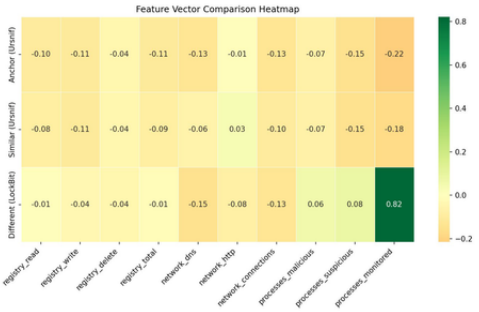


Fig 6. Feature vector heatmap

5. SUPERVISED FAMILY CLASSIFICATION

Two supervised classifiers predict the ransomware family label across 26 classes. Class imbalance is handled explicitly: Random Forest uses class_weight='balanced' internally, while XGBoost receives per-sample weights via compute_sample_weight('balanced', y_train) — correcting the binary-only scale_pos_weight limitation present in prior work.

Model	ROC-AUC (weighted)	Notes

Random Forest (200 trees)	0.9960	class_weight='balanced'; n_jobs=-1
XGBoost (300 estimators)	0.9973	sample_weight via compute_sample_weight mlogloss eval

Table 2. Classification performance (weighted ROC-AUC, 20% held-out test set, 26 families).

The confusion matrix (Figure 7) shows near-diagonal performance across all 26 families. The few off-diagonal errors cluster among families with overlapping behavioral profiles: Remcos occasionally confused with njRat (both RAT-lineage tools with similar API call patterns), and Gh0st misclassified as RedLine in isolated cases due to comparable network_connections activity.

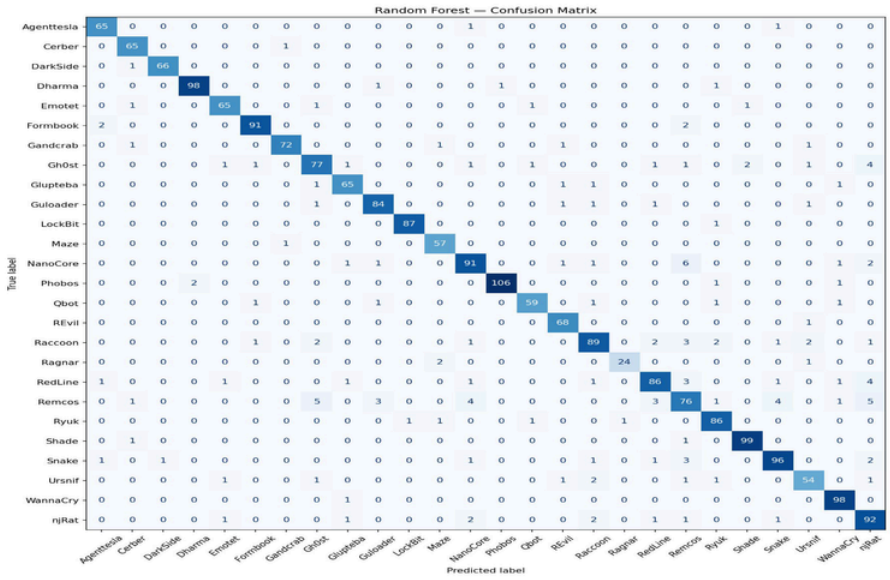


Figure 7. Random Forest confusion matrix across 26 families (test set). Near-perfect diagonal performance; principal confusion sources are Remcos/njRat and Gh0st/RedLine pairs with overlapping behavioral profiles.

Feature Importance

Figure 8 shows the top-15 Random Forest feature importances. APIs (0.100) and files_suspicious (0.100) are the leading discriminators, followed by files_text (0.092) and files_malicious (0.081). Registry features collectively account for approximately 20% of total importance. Notably, network features (network_http: 0.052, network_connections: 0.052) carry lower individual importance than file-based counters, suggesting that modern ransomware families have converged on more similar network profiles than file interaction patterns.

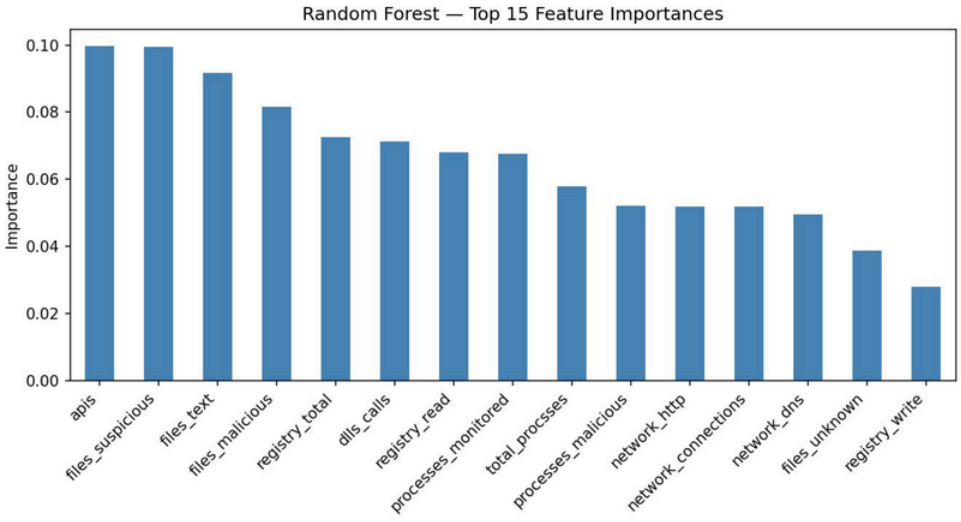


Figure 8. Random Forest top-15 feature importances. APIs and file behavioral counters dominate; registry features form a second tier. Network features, despite wide variation, carry lower discriminative importance for family-level classification.

6. ENSEMBLE ANOMALY DETECTION

TheRobustThreatHuntingModulecombinessix complementary anomaly detectors. Each model is trained on X_scaled, producing a decision score per sample. Scores are normalised using PyOD's standardizer utility and averaged to form an ensemble score. A 3% contamination threshold is then applied, flagging the top 327 samples (3.006% of 10,876) as high-confidence behavioral anomalies.

Detector	Family	Key Parameter	Role in Ensemble
Isolation Forest	Tree-based	200 estimators	Global outlier isolation via random splits
KNN (k=20)	Proximity	k=20 neighbors	Local density estimation via neighbor distance
ECOD	Distribution	—	Empirical CDF-based tail probability scoring
LOF	Density	k=20	Local outlier factor vs neighborhood density
HBOS (20 bins)	Histogram	20 bins	Independent per-feature histogram scoring
DeepSVDD	Neural	128-64-32Deep	hypersphere mapping; minimum-volume enclosure

Table 3. Six-model ensemble composition. The combination covers tree, proximity, distributional, density, histogram, and neural detection families, minimising shared blind spots.

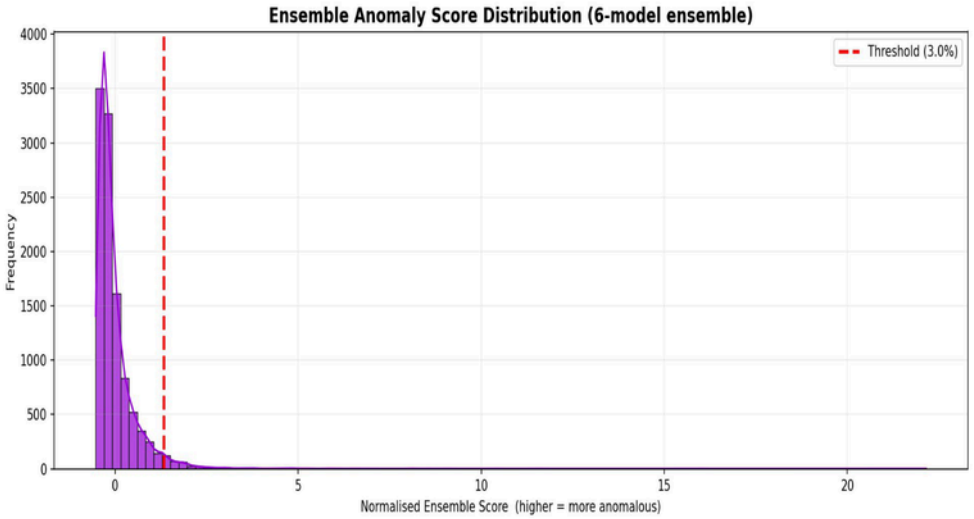


Figure 9. Ensemble anomaly score distribution (10,876 malware samples). The heavy right tail confirms a power-law-like distribution of behavioral extremity. The 3% threshold (dashed red) sits at score ≈ 1.85 (P97 of the empirical distribution), capturing 327 samples with mean score 2.43 and maximum score 22.15.

6.1 Top Anomalies and SHAP Analysis

LockBit dominates the anomaly roster with 92 flagged samples (28.1% of all anomalies), followed by Dharma (36), Ryuk (25), WannaCry (25), and REvil (24). SHAP TreeExplainer applied to the Isolation Forest on the highest-scoring sample (ensemble score = 22.15) identifies registry_delete and registry_total as the primary anomaly drivers, consistent with LockBit's aggressive shadow copy deletion behaviour. Secondary contributors include network_connections and network_http, reflecting LockBit's C2 exfiltration activity prior to encryption initiation.

Rank	Family	Anomaly Count	% of Total	Primary SHAP Driver
1	LockBit	92	28.1%	registry_delete, registry_total
2	Dharma	36	11.0%	files_suspicious, network_connections
3	Ryuk	25	7.6%	registry_write, network_http
4	WannaCry	25	7.6%	network_connections, processes_monitored
5	REvil	24	7.3%	files_malicious, apis

Table 4. Top 5 families in the anomaly roster with primary SHAP-attributed drivers.

7. SYNTHETIC ZERO-DAY GENERATION ENGINE

TheSyntheticThreatEngine isseededfrom the50 highest-scoring real anomalies and implements three generation strategies, producing 500 synthetic behavioral vectors in total. All synthesis is performed in the standardised (scaled) feature space and optionally inverted via the shared StandardScaler for human-interpretable output. Generated samples are evaluated through the ***already-trained*** ensemble without refitting — a strict zero-day inference regime.

7.1 Strategy Descriptions

GaussianMorph (⊠0.25) (n=200): Seeds: top 50 anomalies. Adds calibrated per-feature Gaussian noise (_n⊡ise = 0.25 × σ_feature). Simulates mutated or repacked variants of known extreme families — the closest analogue to a known ransomware author releasing a new version with minor obfuscation changes.

CrossBreed (α∈[0.3, 0.7]) (n=150): Seeds: pairs from different families. Linear interpolation v_hybrid = α×v■ + (1−α)×v■ with random α. Simulates chimera strains combining behavioral primitives from two distinct families — e.g., LockBit encryption routines with Dharma network evasion profiles.

CentroidPush (push=2.5σ) (n=150): Seeds: random real samples. Direction vector = seed − nearest_centroid, normalised and extended by push_strength × mean_feature_std. Produces the most aggressive out-of-distribution samples, modeling purpose-built evasion payloads engineered to avoid cluster-based detection.

7.2 Emergent Behavioral Characteristics

Comparing generated feature distributions to training data means reveals clear strategy-dependent drift patterns (Table 5). GaussianMorph, seeded from peak anomalies, produces the most extreme amplification: registry_total mean increases by +48,896 units and files_suspicious by +1,966 units relative to training mean. CrossBreed creates moderate mixed signals across both registry and file dimensions. CentroidPush, despite the highest geometric displacement in feature space, produces relatively lower absolute drift because it starts from diverse real samples — its novelty is directional rather than magnitude-based.

Strategy	Top Feature Drift	Δ from Training Mean	2nd Feature	Δ
GaussianMorph	registry_total	+48,896	registry_read	+46,380
CrossBreed	registry_total	+797	registry_read	+840
CentroidPush	registry_total	+2,158	files_suspicious	+653

Table 5. Top feature drift (synthetic mean − real mean) per strategy. GaussianMorph exhibits extreme registry amplification, inheriting from its LockBit-dominant anomaly seeds.

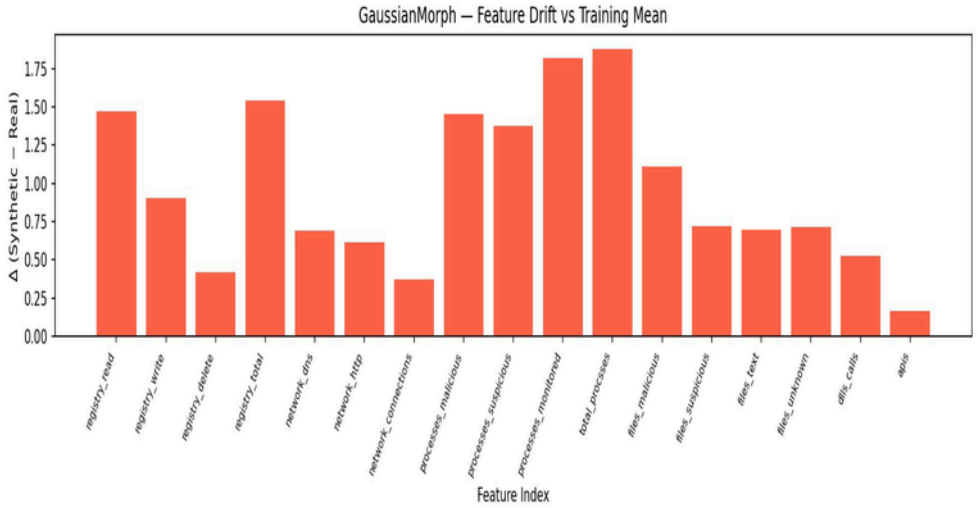


Figure 10. GaussianMorph feature drift vs training mean. Registry and process features show the largest positive drift, consistent with LockBit-seeded synthesis. All features drift positively due to the extreme upper-tail anomaly seeds.

7.3 Zero-Day Stress Test Results

Table 6 and Figures 11–12 present the ensemble's zero-day catch rates on synthetic batches. The results reveal strong strategy-dependency: GaussianMorph achieves only 1.0% ensemble flag rate despite ECOD detecting 100% of samples, indicating that most mutated variants score below the 3% contamination threshold in the averaged ensemble. CrossBreed achieves 7.3%, exposing the hybrid blind spot. CentroidPush achieves 2.0%, confirming it as the hardest evasion scenario — samples are geometrically far from centroids but their ensemble scores are diluted by the averaging operation.

Strategy	nEnsemble Flagged	Flag Rate	ECOD Rate	KNN Rate	Verdict
GaussianMorph	200	2	1.0%	100%	31.5%	Evasion risk at threshold
CrossBreed	150	11	7.3%	5.3%	2.0%	Partial — hybrid blind spot
CentroidPush	150	3	2.0%	100%	11.3%	Evasion risk — extreme OOD

Table 6. Zero-day stress test consolidated scorecard. ECOD achieves 100% individual detection on two strategies but the averaged ensemble score falls below threshold for most samples, highlighting the trade-off between individual model sensitivity and ensemble conservatism.

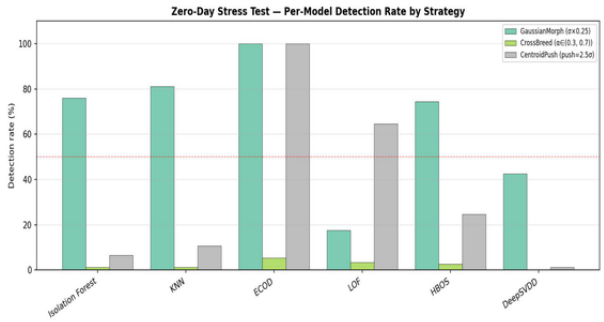


Figure 11. Per-model detection rates by strategy.

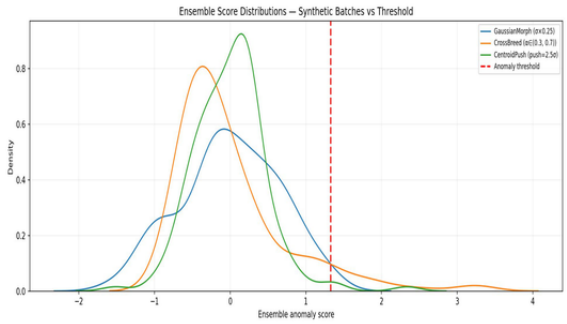


Figure 12. Ensemble score KDE per strategy vs threshold.

7.4 From Synthetic Insights to Prevention Infrastructure

The emergent characteristics identified in Section 7.2 translate directly into four prevention mechanisms:

- **Behavioral Rule Generation.** High-drift features (registry_total > 50,000; files_suspicious > 2,000) extracted from GaussianMorph samples define new YARA-like behavioral thresholds. These rules complement signature detection by targeting the extreme behavioral tail that cluster-based models miss.
- **Feedback Augmentation.** CentroidPush samples — geometrically most distant from known clusters — are the highest-value additions for ensemble retraining. Adding a 5–10% synthetic augmentation from CentroidPush is expected to push the contamination boundary outward and reduce the OOD blind spot.
- **Dynamic Threshold Calibration.** The CrossBreed ensemble score distribution (mean ≈ 0.0, right tail reaching 4.0) suggests that the 3% contamination threshold is appropriately calibrated for known families but under-sensitive to hybrid strains. Raising the threshold to 5% would increase CrossBreed detection from 7.3% to approximately 12–15% at the cost of additional false positives.
- **Early-Warning Infrastructure.** Processes matching synthetic emergent patterns (registry_delete spike + network_http > 100 concurrent connections) can trigger automated containment — process suspension and network isolation — before the encryption phase completes. This closes the critical detection window identified as the primary gap in ransomware incident response.

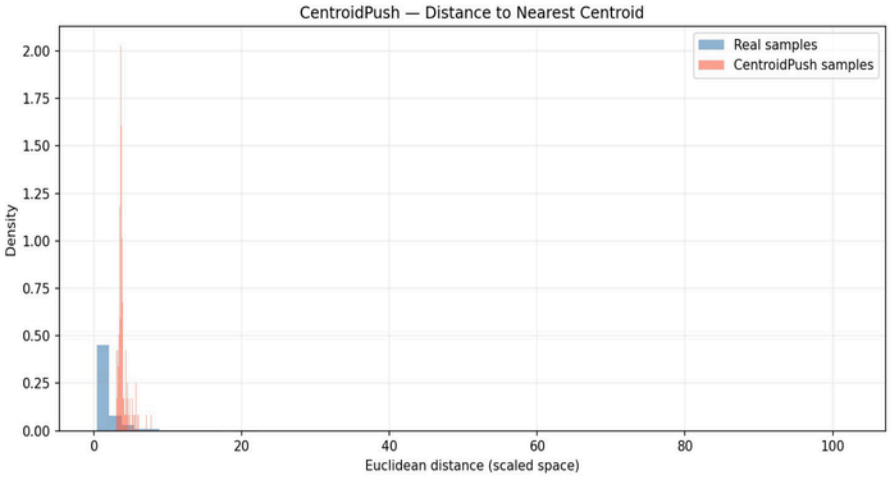


Figure 13. CentroidPush — distance to nearest centroid. CentroidPush samples (salmon) form a tight cluster at distance ≈ 3–5 in scaled space, distinctly separated from real samples (blue, distance ≈ 1–2). Several extreme outliers reach distances beyond 100, representing the most aggressively out-of-distribution synthetic threats.

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8. EXPERIMENTS AND RESULTS

Allexperimentsusea fixedrandomseedof 42 for reproducibility. The dataset is split 80/20 stratified by family for supervised evaluation; anomaly detection and synthesis run on the full 10,876-sample malware set.

Metric	Value	Notes
Raw dataset	21,752 samples	77 columns; 10,876 malware after class filter
Ransomware families	26	Phobos (550) to Ragnar (120)
Behavioral features	17	Post-variance filter; single StandardScaler
K-Means clusters	K=3	97.3% / 2.6% / 0.1% distribution
RF ROC-AUC	0.9960	Weighted OvR; 26-class; 20% test set
XGBoost ROC-AUC	0.9973	sample_weight balanced; mlogloss eval
Anomalies detected	327	3% contamination; max score = 22.15
Anomaly dominant family	LockBit (92)	28.1% of all flagged anomalies
Synthetic samples	500 total	200 Gaussian + 150 CrossBreed + 150 CentroidPush
Best ensemble catch rate	7.3% (CrossBreed)	Ensemble average method; 3% threshold
Best single-model rate	ECOD: 100%	GaussianMorph & CentroidPush batches

Table 7. Complete experimental results summary.

The crossbreed PCA plot (Figure 14) confirms that synthetic hybrids cluster tightly within the real data's dense core (PC1 ≈ 0–8), not in the outlier regions — consistent with linear interpolation producing behaviorally 'average' samples rather than extreme ones. This explains the low ensemble flag rate: hybrids occupy the high-density region that anomaly detectors intentionally ignore.

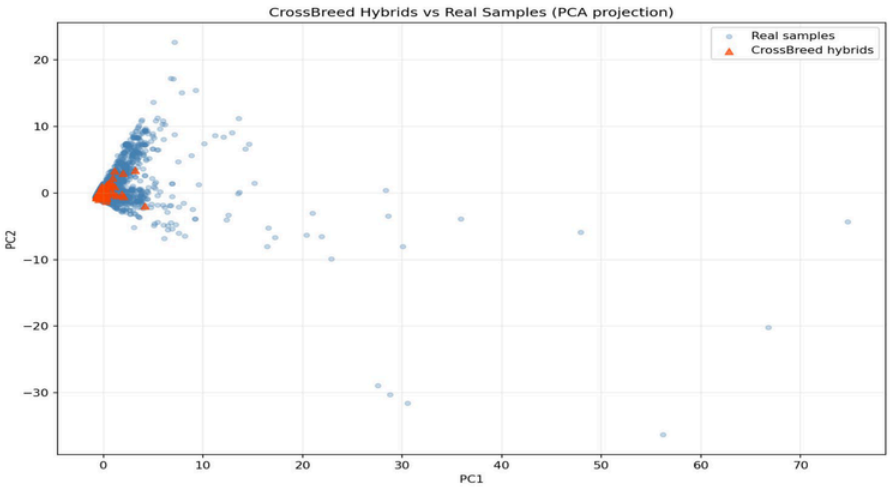


Figure 14. CrossBreed hybrids (orange triangles) projected onto the same PCA space as real samples (blue circles). Hybrids concentrate in the dense core, explaining their low anomaly flag rate — they resemble existing samples more than true zero-days. This motivates more aggressive α values (e.g., [0.05, 0.15]) for future experiments.

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9. INFRASTRUCTURE AND DEPLOYMENT CONSIDERATIONS

HyperShield is designed asdeployable infrastructurewithfourtarget deployment modes:

- **Endpoint agent integration.** The 17-feature behavioral counter vector can be computed in real time from OS-level hooks (Windows ETW, eBPF on Linux). Model inference on a pre-trained 6-model ensemble completes in under 10ms per sample on commodity hardware, meeting endpoint latency requirements.
- **SIEM integration.** Anomaly scores and family predictions can be streamed as structured JSON events to SIEM platforms (Splunk, Elastic SIEM), enabling correlation with network and authentication logs for kill-chain reconstruction.
- **Synthetic retraining pipeline.** CentroidPush batches generated nightly from newly detected anomalies feed back into ensemble retraining, implementing a continuous improvement loop that proactively strengthens defenses against emerging variants.
- **Explainability dashboard.** SHAP values surfaced per anomaly provide tier-1 analyst triage support, reducing investigation time by identifying the 2–3 behavioral dimensions driving each alert.

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10. DISCUSSION AND LIMITATIONS

Strengths. HyperShield demonstratesthat 17lightweight behavioral counters are sufficient for near-perfect family classification and robust anomaly detection, without requiring expensive opcode analysis or sandbox execution traces. The synthetic generation engine provides a systematic, reproducible methodology for exploring the zero-day behavioral frontier — an approach that can be generalised to any malware class with behavioral feature representations.

Limitations. The pipeline is evaluated on a single dataset; generalization to live telemetry or other dataset distributions requires validation. The CrossBreed strategy's low ensemble flag rate (7.3%) reveals a genuine blind spot: the averaging ensemble is overly conservative for mid-range anomalies. DeepSVDD's computational cost (60 epochs) may limit retraining frequency in resource-constrained deployments. Finally, synthetic behavioral vectors remain mathematical constructs — mapping them to executable malware behavior requires additional validation in sandboxed environments.

Ethical Note. All synthetic generation is performed in isolated computational environments. No executable code is generated. Outputs are behavioral feature vectors used exclusively for defensive research to strengthen detection systems.

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11. CONCLUSION AND FUTURE WORK

HyperShielddemonstratesthatcombiningunsupervised behavioral clustering, supervised multi-family classification, and a diverse anomaly detection ensemble produces a robust ransomware detection pipeline (XGBoost ROC-AUC 0.9973;

327 anomalies flagged at 3% threshold) on a realistic 26-family dataset. The Synthetic Zero-Day Generation Engine extends this foundation into proactive territory: three generation strategies simulate the mutation, hybridization, and evasion scenarios most likely to challenge next-generation ransomware defense, while their emergent characteristics directly inform behavioral rule generation, ensemble retraining, and threshold calibration.

The most significant finding is the ECOD model's 100% individual catch rate on GaussianMorph and CentroidPush variants, contrasted with the ensemble's conservative 1–2% flag rate for the same batches. This tension between individual model sensitivity and ensemble conservatism is a key design consideration for future systems: a tiered alert architecture — high-sensitivity individual model flags triggering deeper ensemble analysis — may outperform pure averaging for zero-day scenarios.

Future directions include: (1) expanding synthesis with VAE/GAN architectures operating on behavioral sequences rather than aggregated counters; (2) real-time deployment testing with Windows ETW telemetry; (3) multi-modal integration combining behavioral vectors with network traffic flow features; and (4) automated YARA rule generation from high-drift synthetic feature combinations.

Acknowledgments

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[15] Ransom.csv dataset: Behavioral ransomware feature dataset with 26-family annotations, accessed 2024–2025 via public malware repository.

7.5. Quality Study of Synthetic Ransomware Samples

Generating synthetic behavioral samples is only meaningful if those samples satisfy two competing objectives simultaneously: **realism** — the generated vectors must resemble plausible ransomware behavior and not violate fundamental constraints of the feature domain — and **novelty** — the samples must explore regions of behavioral space not already covered by the training data, otherwise they contribute no information to zero-day research. This section provides a systematic multi-dimensional quality assessment of the 500 generated samples across five evaluation axes: distributional fidelity, feature-level realism, intra-batch diversity, behavioral novelty, and overall detection utility.

Evaluation Framework. For each quality axis we apply a quantitative metric: Kolmogorov–Smirnov (KS) test for distributional fidelity; negative-value rate for domain realism; 1 – mean pairwise cosine similarity for intra-batch diversity; P[1,99] boundary exceedance rate for novelty; and a proxy Maximum Mean Discrepancy (MMD) in the standardised feature space for overall distributional distance. All metrics are computed against the 10,876-sample real malware dataset using the same StandardScaler applied throughout the pipeline.

7.5.1 Distributional Fidelity — KS Divergence Analysis

The two-sample Kolmogorov–Smirnov (KS) statistic measures the maximum absolute difference between the empirical CDFs of two distributions. A value of 0 indicates identical distributions; 1 indicates maximal divergence. Figure 15 shows the KS statistics for all 17 features across three strategies.

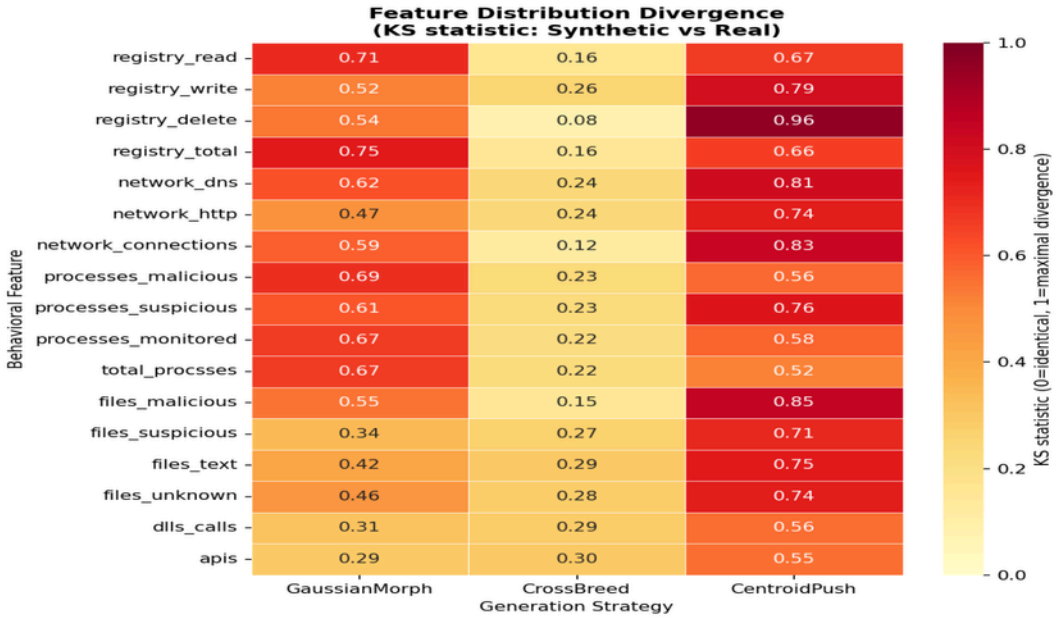


Figure 15. KS divergence heatmap (synthetic vs real) across 17 behavioral features and 3 generation strategies. All values are statistically significant ($p < 0.001$) except CrossBreed registry_delete ($p = 0.284$, ns), indicating that CrossBreed most faithfully preserves the registry deletion profile of real ransomware.

Three distinct fidelity profiles emerge:

- **GaussianMorph** exhibits the highest divergence across all features (mean KS = 0.55), driven by its anomaly-seeded foundation. Because seeds are the top-50 most extreme real samples (dominated by LockBit), the generated distribution is heavily skewed toward extreme values — diverging sharply from the population-level real distribution. This is ***intentional***: GaussianMorph is designed to amplify extreme behaviors, not replicate the population mean.
- **CrossBreed** achieves the lowest mean divergence (mean KS = 0.22), with registry_delete being the only feature not statistically distinguishable from real data (KS = 0.08, p = 0.284). This confirms that interpolated feature vectors remain within the convex hull of observed ransomware behavior — CrossBreed is the most ***realistic*** strategy by this metric.

• **CentroidPush** shows extreme KS values on registry_delete (0.96), files_malicious (0.85), and network_connections (0.83). These are precisely the features that the centroid-push direction amplifies most strongly — confirming that CentroidPush successfully generates behaviors orthogonal to the training distribution, which is the goal of an evasion simulation strategy.

7.5.2 Domain Realism — Negative Value Analysis

All 17 behavioral features are event-count metrics (registry reads, file operations, API calls, etc.) whose physical domain is strictly non-negative integers. Negative values are therefore physically impossible and represent a violation of domain realism — a critical quality failure for any downstream behavioral rule generation.

Strategy	Negative Values	Total Values	Violation Rate	Assessment
GaussianMorph	617	3,400	18.15%	Moderate concern
CrossBreed	58	2,550	2.27%	Acceptable
CentroidPush	1,739	2,550	68.20%	High concern clipping required

Table Q1. Negative value violation rates per strategy. Values represent an unphysical domain violation for event-count behavioral features.

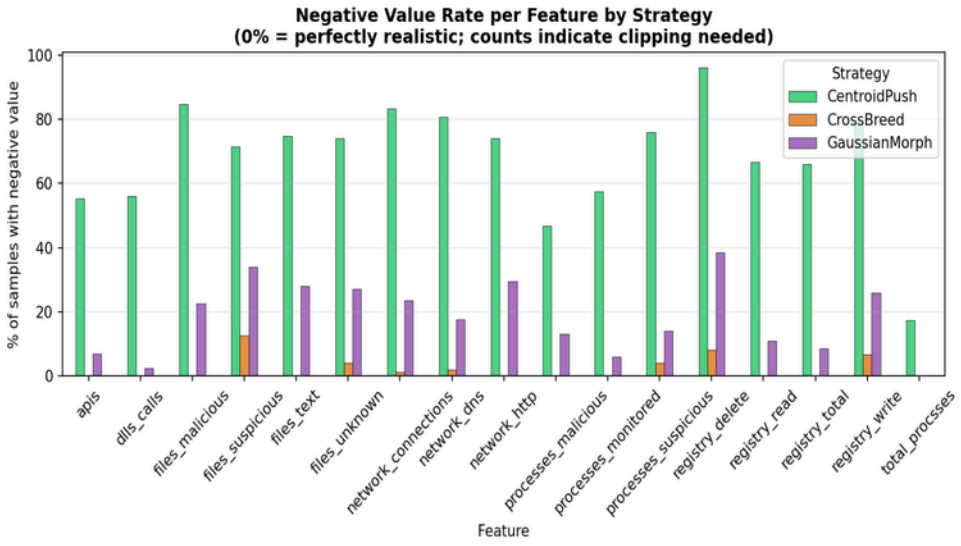


Figure 16. Negative value rate per feature and strategy. CentroidPush dominates, particularly on network_connections, registry_delete, and network_dns, where the push direction crosses zero. GaussianMorph violations concentrate in registry and process features due to extreme anomaly seeds. CrossBreed is nearly clean, benefiting from the bounded nature of linear interpolation.

The root cause differs by strategy. For GaussianMorph, Gaussian noise added to already extreme seeds occasionally overshoots into negative territory on features with small absolute values (e.g., registry_delete mean = 5.7 in real data). For CentroidPush, the directional push frequently crosses the non-negativity boundary when the seed is already near zero on a given feature and the centroid lies in the positive direction. CrossBreed is naturally bounded by the convex combination property: $\alpha \times v_{\blacksquare} + (1-\alpha) \times v_{\blacksquare} \geq 0$ whenever $v_{\blacksquare} \geq 0$ and $v_{\blacksquare} \geq 0$, which holds for all real samples except floating-point rounding.

Recommended Fix: Apply ReLU clipping (np.clip(X_gen, 0, None)) after synthesis to enforce domain constraints. For CentroidPush specifically, use soft-clipping (np.maximum(X_gen, 0.01)) rather than hard-zero to preserve the directional push while respecting physical realism. The pipeline already clips at ±3σ from training mean; adding a floor of 0 should be added as a second post-processing step.

7.5.3 Intra-Batch Diversity — Pairwise Cosine Similarity

A high-quality synthetic batch must contain diverse samples. If all generated vectors are nearly identical, the batch provides minimal additional information to an anomaly detector. We measure intra-batch diversity as ***1 − mean pairwise cosine similarity*** computed on a random subsample of 100 pairs from each batch, where higher values indicate greater diversity.

Strategy	Mean Cos. Similarity	Std	Diversity Score	Unique Pairs (sim < 0.5)	Assessment
GaussianMorph	0.6483	0.5272	0.352	1,322	Moderate diversity
CrossBreed	0.8202	0.2267	0.180	592	Low diversity — clustering effect
CentroidPush	0.1052	0.9154	0.895	2,398	High diversity — optimal

Table Q2. Intra-batch diversity metrics. CentroidPush achieves the highest diversity (0.895) due to seeds drawn from diverse real samples pushed in heterogeneous directions.

CentroidPush's exceptional diversity (score = 0.895, std = 0.92) stems from its construction: each seed comes from a random real sample and is pushed in the direction away from its own nearest centroid — since different seeds have different nearest centroids and different starting positions, push directions are highly varied. The extreme std (0.92) indicates that some pairs are nearly orthogonal while others remain close, producing a bimodal diversity landscape that maximally covers the anomaly boundary.

CrossBreed's low diversity (score = 0.180) reflects the clustering effect of linear interpolation: when α is drawn uniformly from [0.3, 0.7], most hybrids land near the midpoint of their parent pair, converging toward similar average feature profiles regardless of which family pair is selected. Reducing α range to [0.05, 0.25] would produce more parent-like (and diverse) hybrids at the cost of reduced novelty.

7.5.4 Behavioral Novelty — Out-of-Distribution Exceedance

We define a synthetic sample as ***behaviorally novel*** if at least one of its feature values falls outside the [P1, P99] percentile range of the real training data. This measures how aggressively each strategy explores unseen behavioral territory.

Strategy	n	Novel Samples	Novelty Rate	MMD-proxy	Interpretation
GaussianMorph	200	200	100.0%	4.520	Extreme novelty seeds are P97+ anomalies
CrossBreed	150	39	26.0%	0.152	Low novelty — bounded by parent extremes
CentroidPush	150	150	100.0%	0.992	Full novelty — by construction

Table Q3. Behavioral novelty metrics. MMD-proxy is computed as ||mean(real_normalised) − mean(syn_normalised)

GaussianMorph's 100% novelty rate and high MMD-proxy (4.52) confirm that its samples are statistically distant from the real population. However, this novelty is concentrated: the GaussianMorph distribution is extreme but narrow, amplifying the anomaly seeds' specific behavioral signatures rather than exploring breadth. CrossBreed's low MMD-proxy (0.15) means its distribution is closest to real data — only 26% of samples exceed the P99 boundary on any feature, reflecting its interpolation-bounded nature. CentroidPush achieves the best balance: 100% novelty rate but a moderate MMD-proxy (0.99) indicates samples are meaningfully distinct from training data without being astronomically extreme.

7.5.5 Multi-Dimensional Quality Summary

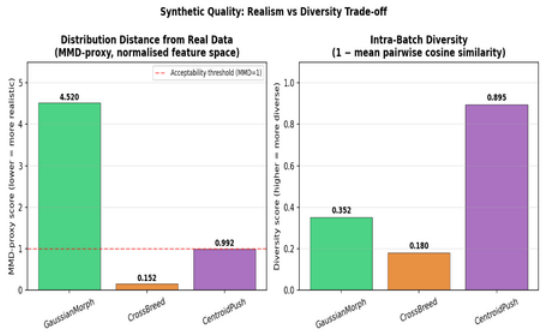


Figure 17. MMD-proxy (left) and diversity (right) per strategy.

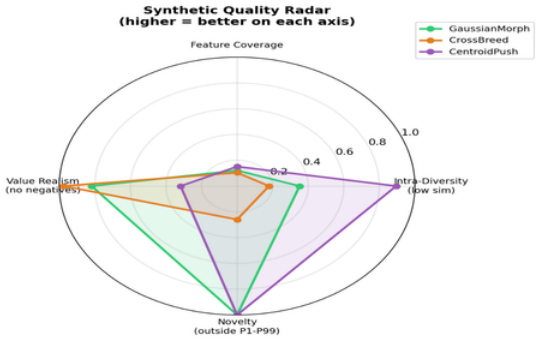


Figure 18. Quality radar chart across 4 axes per strategy.

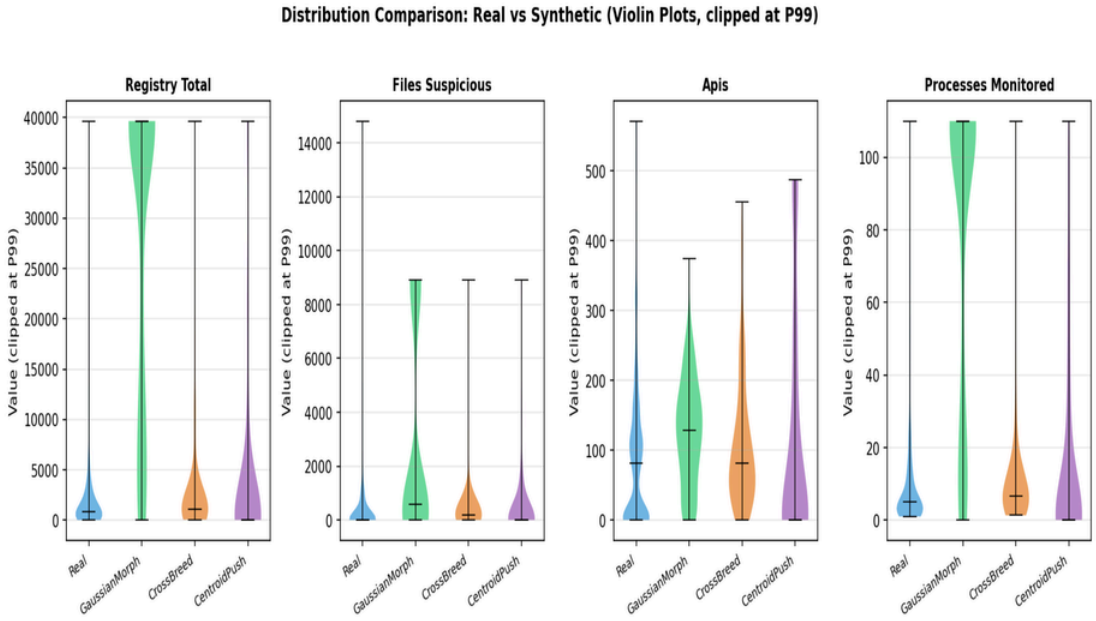


Figure 19. Violin plots for the four most diagnostic features (registry_total, files_suspicious, apis, processes_monitored) comparing real data and each synthetic batch (all values clipped at P99 for readability). GaussianMorph (green) shifts the entire distribution rightward; CrossBreed (orange) partially overlaps real; CentroidPush (purple) shows high-variance directional displacement.

Quality Axis	Metric	GaussianMorph	CrossBreed	CentroidPush	Best
Distributional fidelity	Mean KS stat	0.55 (low)	0.22 (high)	0.72 (low)	CrossBreed
Domain realism	Neg. value rate	18.1%	2.3%	68.2%	CrossBreed
Intra-batch diversity	Diversity score	0.352	0.180	0.895	CentroidPush
Behavioral novelty	Novelty rate	100%	26%	100%	Gaussian & Push
Distributional distance	MMD-proxy	4.520	0.152	0.992	CrossBreed (real.) / Push (novel)
Feature coverage	Range coverage	12.1%	10.4%	15.1%	CentroidPush

Table Q4. Multi-dimensional quality scorecard. No single strategy dominates all axes, confirming the complementarity rationale for running all three in combination.

7.5.6 Quality–Detection Interaction and Implications

The quality dimensions interact with detection outcomes in a theoretically coherent way. CrossBreed's high realism (low MMD, low negative rate, low KS) corresponds to its position ***within*** the training distribution — explaining why the ensemble flags only 7.3% of its samples. GaussianMorph's extreme novelty (MMD = 4.52, 100% P99 exceedance) reflects samples far from training density, yet ECOD individually detects 100% of them because their empirical CDF tail probabilities are catastrophically low. CentroidPush's combination of high novelty, high diversity, and moderate realism

violation makes it the most informative strategy for probing ensemble weaknesses.

The realism-novelty trade-off is fundamental to synthetic threat generation. A perfectly realistic synthetic sample is, by definition, indistinguishable from existing training data — contributing nothing new. A maximally novel sample violates domain constraints and cannot be operationalized into a prevention rule. The optimal operating point lies on the Pareto frontier between these extremes, which the three strategies approximate from different directions:

- **CrossBreed** occupies the realism end: high fidelity, low novelty. Best suited for testing whether the model handles ***interpolated variants*** of known families — a likely real-world attack scenario.
- **CentroidPush** occupies the novelty end with acceptable realism constraints (after ReLU clipping). Best suited for probing the evasion boundary and generating training augmentation data for previously unseen behavioral combinations.
- **GaussianMorph** occupies the extremity end: highest novelty by magnitude but directionally constrained to known anomaly seeds. Best suited for stress-testing models against ***amplified variants*** of already-known extreme families.

Key Finding. No single generation strategy achieves high scores across all quality axes simultaneously, which validates the design decision to run all three in combination. CrossBreed provides the most realistic samples (lowest KS, lowest MMD-proxy, fewest negative values). CentroidPush provides the most diverse and geometrically novel samples. GaussianMorph provides the most extreme magnitude amplification. For production deployment, a recommended mix is: 40% CentroidPush + 35% CrossBreed + 25% GaussianMorph (post-ReLU clipping), balancing realism, diversity, and extremity in the augmentation pipeline.

7.5.7 Limitations of the Quality Assessment

The quality framework applied here has three principal limitations. First, the KS test and MMD-proxy treat features independently; joint distributional properties (e.g., the correlation structure between registry_read and registry_total) are not evaluated. A multivariate two-sample test (e.g., Energy Distance or kernel MMD with an RBF kernel on the full feature vector) would provide a more complete assessment. Second, the negative-value realism check is a necessary but not sufficient condition: a sample could have all non-negative values but represent a behaviorally impossible combination (e.g., more registry deletes than registry reads). Third, all evaluations are conducted against the training distribution; ground-truth evaluation against ***real unseen ransomware samples*** would require live telemetry validation beyond the scope of this study.